DATA ENGINEERING FOR AI
DAY 3

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DATA LAKE VS. DATA WAREHOUSE

DATA LAKE

DATA WAREHOUSE

TRADITIONAL ETL (EXTRACT, TRANSFORM, LOAD)

Small compute, data-movement bottcleck.

MODERN ELT (EXTRACT, LOAD, TRANSFORM)

Large compute, efficient.

BATCH VS STREAMING

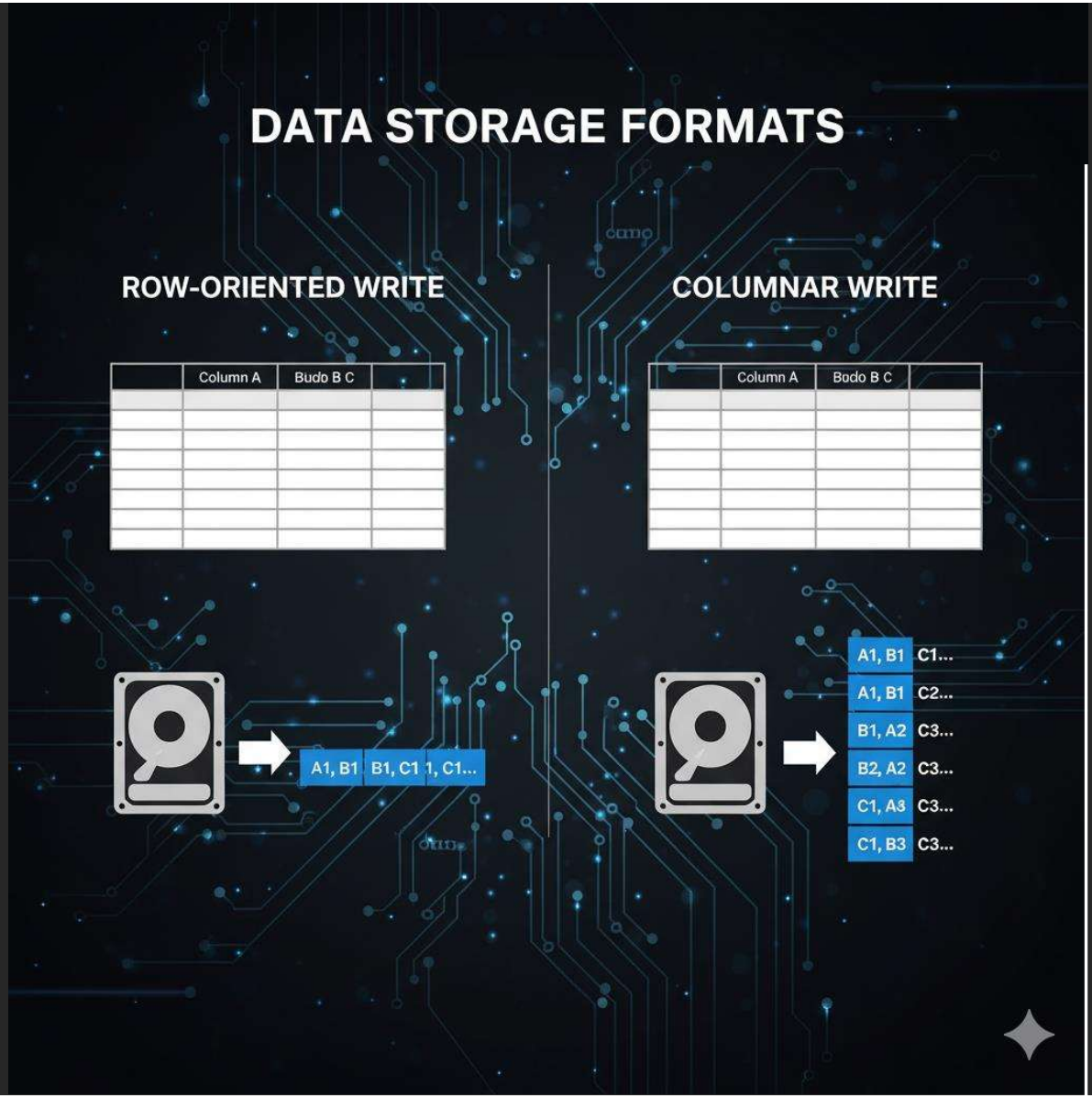
BATCH
(Scheduled Intervals)

Processes data in chunks, perididaly.

STREAMING
(Continuous Flow)

Processes data continouely, as pot arrives.

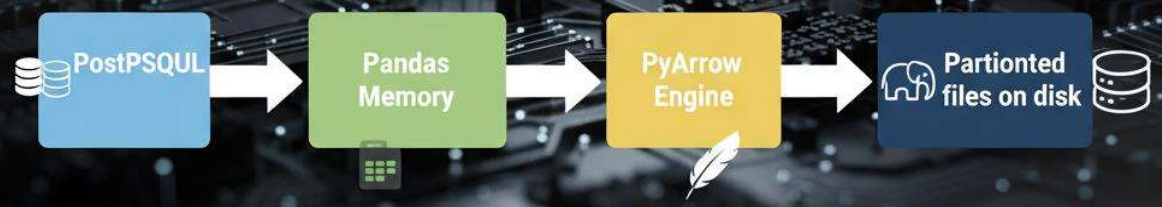
- The stomach (the GPU) has plenty of capacity, but the physical pipe (the database connection) limits the flow to a useless trickle



Row-Oriented vs. Column-Oriented Storage

HuggingFace

DATA FORMAT COMPARISON

CSV (Comma Separated Values)	**PARQUET** (Columnar Format)
FILE SIZE: **10 GB**	**2 GB**
READ TIME: **60 SECONDS**	READ TIME: **2 SECONDS**

DATA EXPORT ARCHITECTURE